

CS3700 Lab 2: Generalization and Specialization

Objective

Apply generalization and specialization to create Entity-Relationship Diagrams (ERDs) using Chen notation.

Instructions

Complete **all four case studies**.

For each case study:

1. Identify whether **generalization** or **specialization** best describes the modeling process. Explain your reasoning.
2. Identify the superclass and subclasses.
3. Identify shared and subclass-specific attributes.
4. Draw an ERD that includes:
 - Entities represented as rectangles.
 - Attributes represented as ovals, with key attributes underlined.
 - Shared attributes attached to the superclass.
 - Subclass-specific attributes attached to the appropriate subclasses.
 - A double line for **total** specialization or a single line for **partial** specialization.
5. Justify your disjoint/overlapping and total/partial choices using the business rules.
6. Answer the discussion question.

Subclasses inherit the superclass's attributes and key. Do not repeat inherited attributes in each subclass.

Case Study 1: Campus Parking

The university currently stores all registered vehicles in a single **Vehicle** entity with these attributes:

- VehicleID – unique identifier
- LicensePlate
- Manufacturer
- Model

The parking office now needs to record **SeatingCapacity** for cars and **EngineDisplacement** for motorcycles.

A registered vehicle cannot be both a car and a motorcycle. The system also registers other vehicles, such as buses, without recording additional type-specific attributes.

Discussion question: Why might adding both new attributes directly to Vehicle be an unsuitable design?

Case Study 2: University Contacts

Two university offices maintain separate records:

Student	Employee
UniversityID	UniversityID
Name	Name
Email	Email
Major	HireDate
AdmissionYear	Salary

The university wants to combine these records into one model. UniversityID uniquely identifies a person and identifies the same person across both offices.

A person may be both a student and an employee. The combined system will contain only people who are students, employees, or both.

Discussion question: How should the model represent someone who is both a student and an employee without duplicating their shared information?

Case Study 3: Library Resources

A library currently uses a single **Resource** entity with these attributes:

- ResourceID – unique identifier
- Title
- PublicationYear

The library tracks only printed books and e-books. Every resource must belong to exactly one of these categories.

Printed books require an **ISBN** and a **ShelfLocation**. E-books require an **AccessURL** and a **FileFormat**.

If a title is available in both formats, each format receives a separate ResourceID.

Discussion question: Does having the same title in both formats make the subclasses overlapping? Explain using the resource identity rule.

Case Study 4: Equipment Inventory

A media lab maintains separate equipment records:

Camera	AudioRecorder
AssetID	AssetID
Manufacturer	Manufacturer
PurchaseDate	PurchaseDate
MegapixelCount	ChannelCount
LensMount	MaximumSampleRate

The lab wants to consolidate shared information. AssetID uniquely identifies a physical device across the entire inventory.

Some hybrid devices qualify as both cameras and audio recorders and require both sets of type-specific attributes. The inventory also includes lighting equipment, for which only the shared attributes are currently recorded.

Discussion question: Where should the model store a hybrid device's Manufacturer and PurchaseDate?

Submission

Submit **one PDF** named `Lastname_Firstname_GeneralizationSpecialization.pdf`.

Organize your submission into four labeled sections. Each section must include:

- Your generalization or specialization decision and explanation.
- The superclass, subclasses, and their attributes.
- A complete, readable ERD.
- Justifications for both subclass constraints.
- Your answer to the discussion question.

Diagrams may be created digitally. All work must be completed individually.

Grading - 40 Points

Each case study is worth **10 points**.

Criterion	Points per case
Generalization/specialization decision and explanation	2
Superclass, subclasses, and attribute placement	3
Correct Chen, including key identification	2
Correct subclass constraints and justifications	2
Discussion response	1